Firewalls

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Network Security Systems

A brief overview of some basic network security systems.

Iptables and Firewalls

From Wikipedia:

> *A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.*
>
> *A firewall typically establishes a barrier between a trusted network and an untrusted network.*

Firewall examples: AWS

```
MyExampleSecurityGroup:
   Type: 'AWS::EC2::SecurityGroup'
   Properties:
     SecurityGroupIngress:
       - IpProtocol: tcp
         FromPort: '22'
         ToPort: '22'
         CidrIp: 130.108.0.0/16
       - IpProtocol: tcp
         FromPort: '1025'
         ToPort: '2048'
         CidrIp: 130.108.0.0/16
       - IpProtocol: -1
         FromPort: '-1'
         ToPort: '-1'
```

Firewall examples: iptables

```
# Generated by iptables-save v1.8.7 on <date>
*filter
:INPUT DROP [0:0]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
-A INPUT -p tcp --dport 22 -j ACCEPT
-A INPUT -p tcp --match multiport --dports 1025:2048 \
    -j ACCEPT
-A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
-A INPUT -p tcp -s 10.0.0.0/16 -j ACCEPT
COMMIT
```

Firewall Types

Packet filters

- **Stateless** filters check basic information like port number, IP address, protocol, etc. of a packet to take action.
- **Stateful** filters maintain records of conections passing through, can determine if a packet is the start of a new connection or a part of an existing connection. In `iptables` the four trackable states are **NEW**, **ESTABLISHED**, **RELATED** and **INVALID**.

Application Layer

- works more like a proxy, can "understand" applications and protocols.
- frequently inspects contents of traffic.

Firewall Policies

Blacklist approach (default-allow)

- All packets are allowed through except those that fit the defined blacklist.
- Pros: flexible, less likely to impact services.
- Cons: unexpected malicious traffic makes it through.
- ***Lab Task 3***

Whitelist approach (default-deny)

- All packets are dropped or rejected unless specifically allowed.
- Pros: safer network
- Cons: unexpected important traffic gets dropped or rejected.
- ***Lab Task 4***

Firewall Rules

Firewalls take one of three actions on a packet.

- *Allow* lets the packet pass through.
- *Drop* prevents the packet from passing but is silent about it.
- *Reject* prevents the packet from passing but notifies the sender that it is being blocked.

ipchains

iptables specfically has multiple chains of rules that apply, here are the 5 predetermined ones:

- PREROUTING are rules applied to all packets before any routing decisions are made.
- INPUT are rules applied on packets that will be delivered locally.
- FORWARD are rules applying to packets that have been routed but are not for locally delivery
- OUTPUT are packets sent by the local machine.
- POSTROUTING again are all packets but after they have been through all previous relevant chains.

`iptables` tables

Each of the chains defined previously can be applied to different tables depending on intent for the packet.

`*filter`

For packets that are destined locally and only to be filtered (allowed through or dropped) rules go in the `filter` table.

`*nat`

For packets destined elsewhere and that will need to be altered there is a `nat` table.

`*mangle`

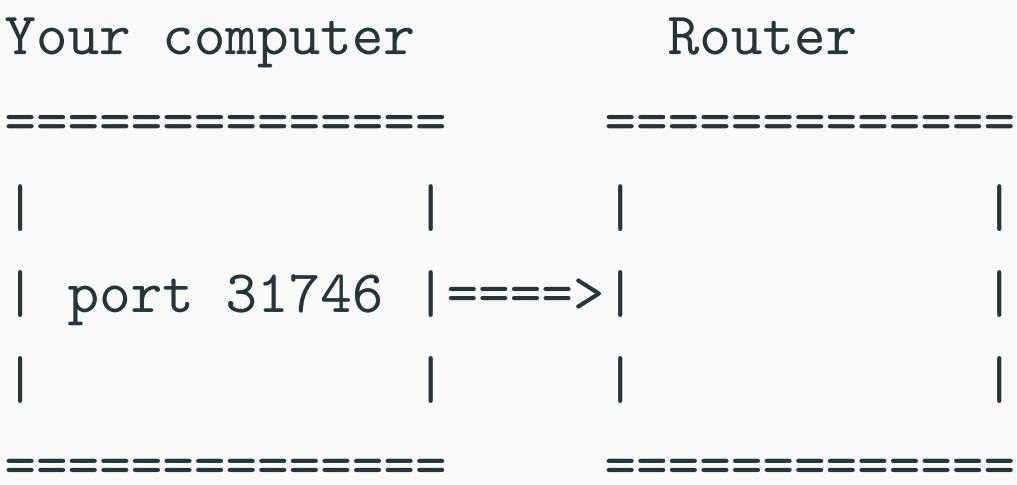
For all other custom packet manipulation there is a `mangle` table.

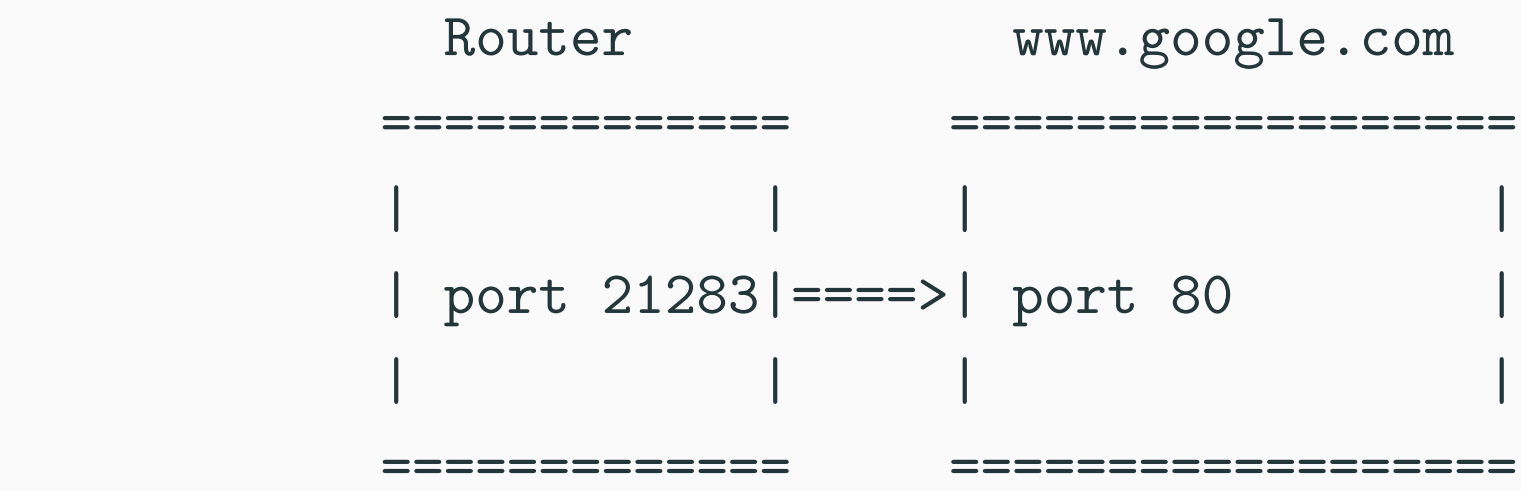
NAT (for l337 gamers and programmers)

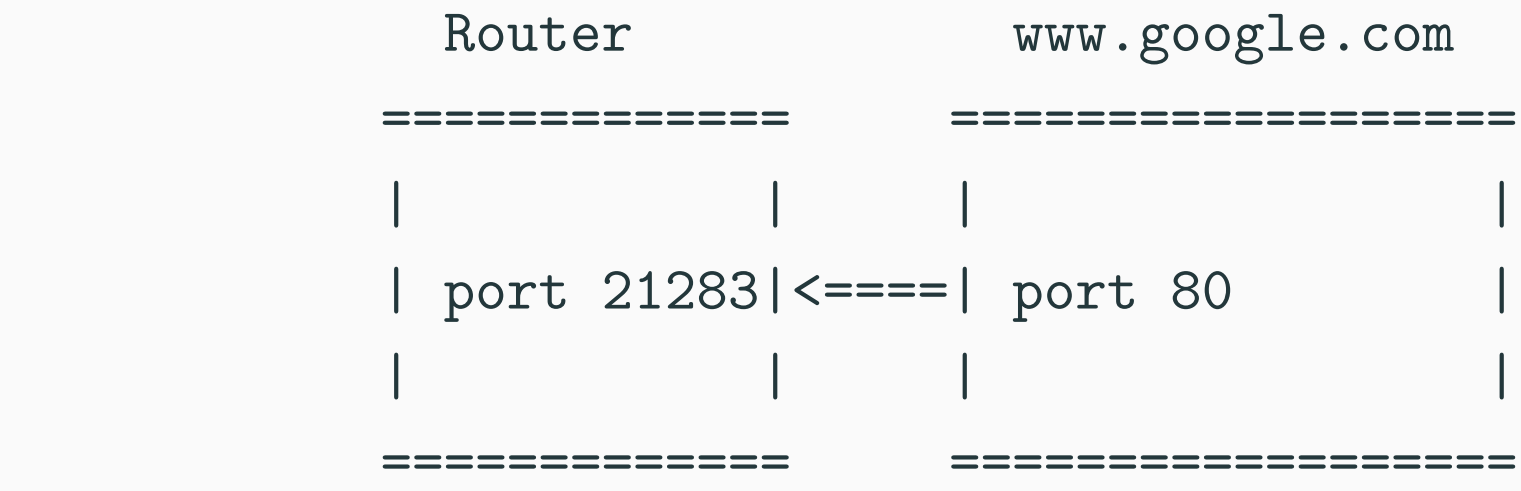
NAT stands for Network Address Translation

This is what allows your non-routable private network `192.168.1.0/24` access to the internet.

Lets say you want to make a connection to google.com:80




Router
www.google.com
port 21283
port 80


Router
www.google.com
port 21283
port 80

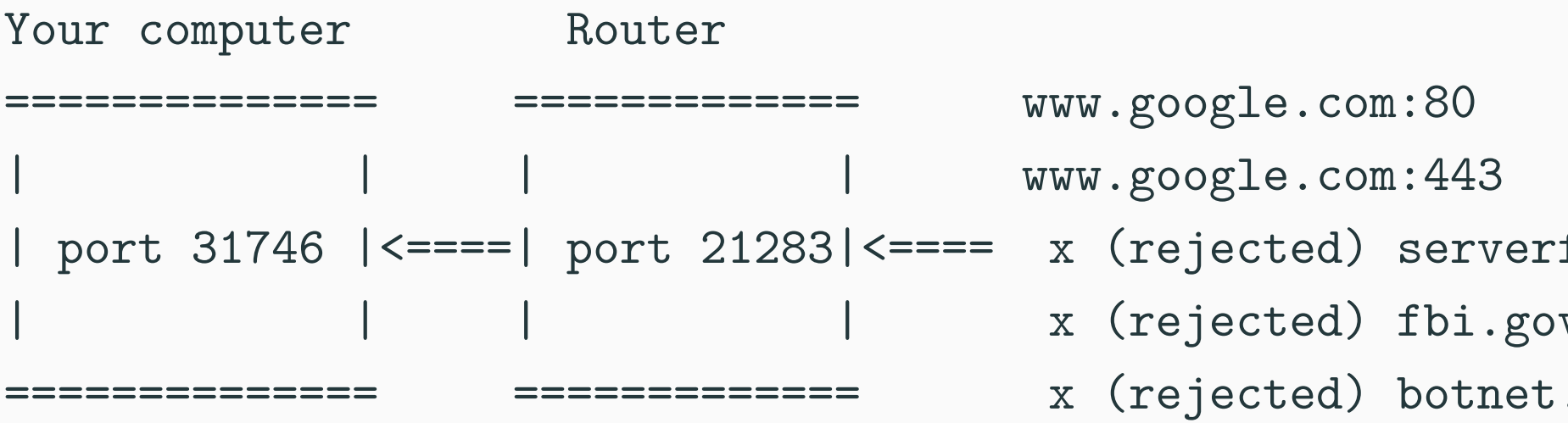
Open NAT

All traffic coming in to your Router on a given port gets sent to your computer.



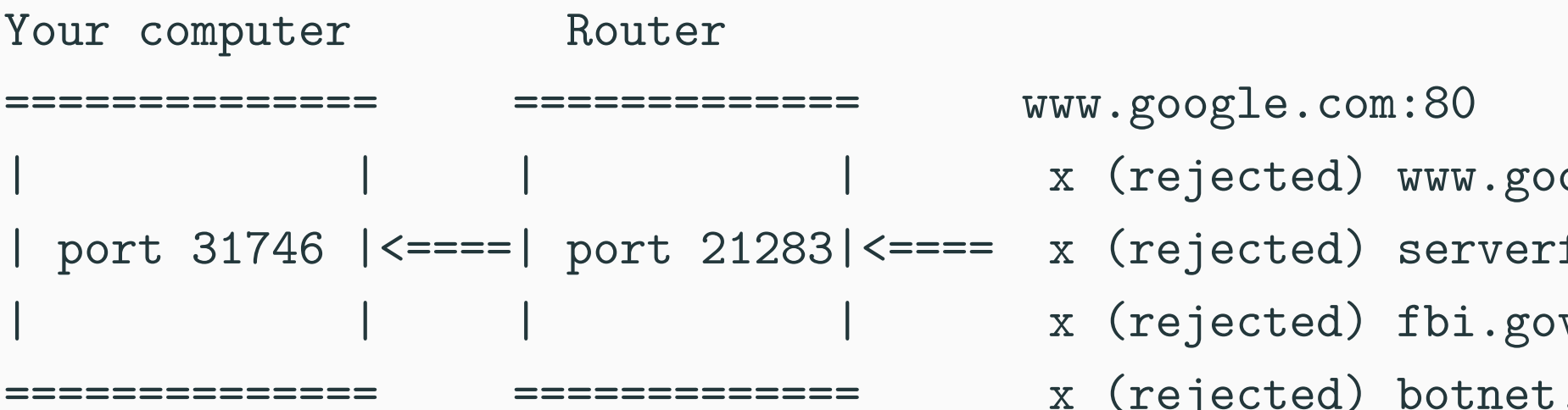
Moderate NAT

Only traffic from a specified *host* gets sent back through to your PC.



Strict NAT

Only responses from the requested *host* **and** *port* are sent back to your pc.



IPtables info (web links)

- iptables guide
- iptables man page
- cool NAT info I used

IPtables notes for the lab

- iptables do not persist through a reboot (so if you lock yourself out you can reboot to get back in)
- iptables can be annoying
- dropping all inbound connections (without any exceptions) will break everything
  - the above will not allow any form of remote communications with the target, which is very annoying ;)